

Branch Specialization in Ordinary Attention Heads

Larger Head-Count Control and Real-Model Role Validation

Codex autonomous research checkpoint

2026-05-24

Why This Checkpoint Exists

User concern:

Maybe family-level modularity looked weak because the toy model had too few heads for a sparse modular distribution.

This round tests that concern before making a stronger claim about real-model validation.

How Many Heads Before?

Previous setting	Heads/layer	Layers	Total head slots
uniform2 / hetero2	2	2	4
uniform4 / hetero4	4	2	8

The concern is valid: 4 or 8 slots is small for testing sparse family-level modularity.

New Larger-Head Setups

Setting	Heads/layer	Layers	Total slots
32-slot larger model	8	4	32
16-slot control	8	2	16

Configs:

- uniform8: [48,48,48,48,48,48,48,48]
- hetero8_unique_spread: [16,24,32,40,48,56,72,96]
- hetero8_unique_extreme: [8,16,24,32,40,48,64,152]

All have total attention dimension 384.

Questions And Metrics

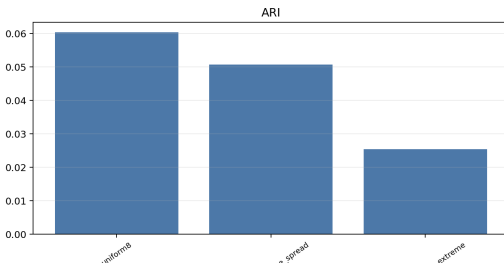
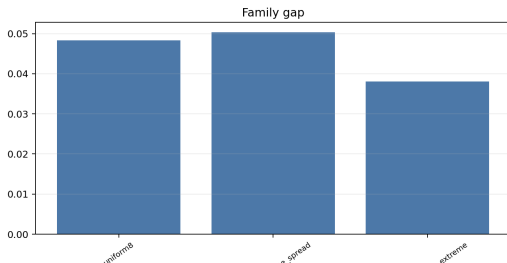
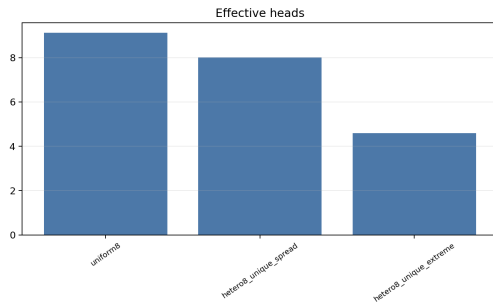
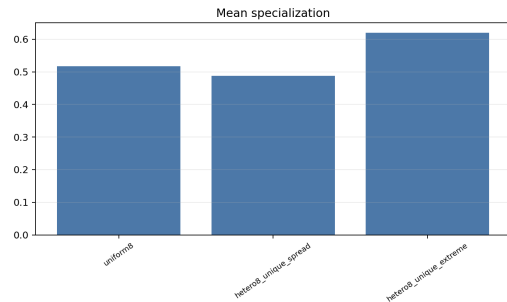
Question	Metric
Structural role affinity	Largest-dimension top rate. Chance for hetero8 is 0.125.
Specialization	Top role mass and effective heads.
Slot-level modularity	Family gap and ARI over exact layer-head slots.
Dimension-level modularity	Same family metrics after collapsing by head dimension.

32-Slot Result

Config	Min acc	Largest top	Spec	Eff heads	Gap	ARI
uniform8	0.996	n/a	0.517	9.117	0.048	0.060
hetero8_spread	0.991	0.50	0.488	8.020	0.050	0.051
hetero8_extreme	0.990	0.72	0.619	4.589	0.038	0.025

Interpretation: structural affinity survives, but family modularity stays weak.

32-Slot Plot



16-Slot Control

The first 1000-step 16-slot run undertrained `uniform8`.

Targeted fix:

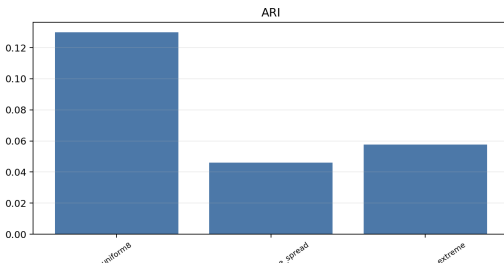
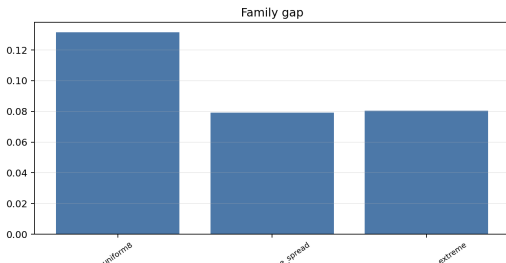
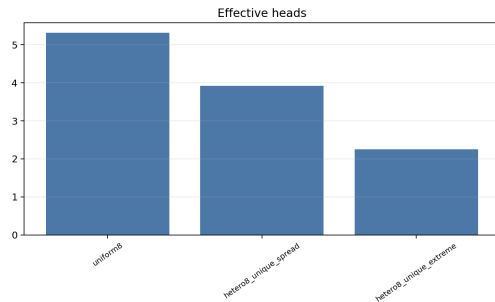
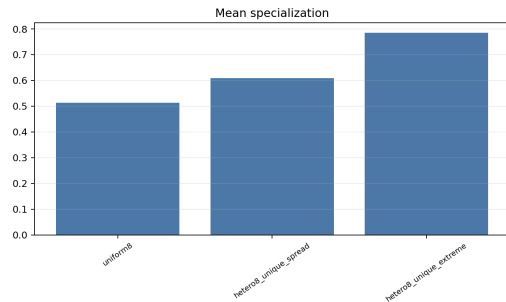
- rerun the same 16-slot grid for 2000 steps;
- use the 2000-step run as the clean comparison.

16-Slot Clean Result

Config	Min acc	Largest top	Spec	Eff heads	Gap	ARI
uniform8	0.978	n/a	0.514	5.314	0.132	0.130
hetero8_spread	0.988	0.62	0.609	3.923	0.079	0.046
hetero8_extreme	0.997	0.76	0.784	2.247	0.081	0.058

Interpretation: hetero improves specialization but does not beat the uniform8 baseline on family modularity.

16-Slot Plot



Answer To The Head-Count Concern

More heads do not rescue the modularity claim.

They preserve the stronger results:

- structural role affinity remains far above chance;
- role-level specialization increases under stronger heterogeneity;
- family-level modularity remains separate and not automatic.

Failure Analysis

Possible factor	Evidence
Too few heads	Less likely after 16-slot and 32-slot controls.
Too much depth fragments exact slots	32-slot gap is weaker than 16-slot gap.
Role difficulty, not family, drives large-head choice	Many unrelated roles go to the largest head.
Ontology family labels are too coarse	Induction and boundary subroles behave differently inside family.
Extreme imbalance causes collapse	Extreme configs specialize strongly but have weak ARI.

Real-Model Probe

Pythia-160M-deduped, revision `step143000`, ordinary attention heads.

The first float16 run produced NaNs in later-layer attention summaries. The float32 rerun had no NaN rows and is the one reported here.

Pythia Attention Role Results

Role	Best layer	Best head	Top spec	Eff heads	Interpretation
repeat_match	0	7	0.834	2.088	strong repeat/induction-like role
repeat_match	6	10	0.725	3.445	second strong repeat layer
previous_token	10	9	0.332	9.169	measurable but distributed
bos	8	6	0.140	9.809	diffuse sink behavior

Existing Causal Pythia Results

Probe	Own top	Random	Excess	Aligned	Same-index
Local copy	2.519	0.229	2.290	2.271	0.488
Natural repeat 8-gram	0.387	0.016	0.372	0.315	0.033

Values are loss deltas from ordinary-head ablations in Pythia-160M seed runs.

- Real ordinary heads clearly support local-copy and repeat/induction roles.
- The real-model evidence validates the role-measurement side of the project.
- It does not test heterogeneous dimensions, because Pythia heads are uniform.
- The current paper claim should separate structural affinity, specialization, and modularity.

Heterogeneous ordinary attention-head dimensions create structural role affinity and usually increase role-level specialization.

Larger-head controls do not show that heterogeneity automatically improves family-level modularity.

Next Step

- ① Keep the larger-head result as a negative/moderating result for modularity.
- ② Add role-difficulty analysis: sequence distance, conflict, target entropy, and largest-head preference.
- ③ Expand weak ontology families: induction, boundary/sink, suppression/conflict.
- ④ Continue real ordinary-head causal probes for repeat, local copy, copy suppression, and IOI.
- ⑤ Do not claim full functional modularity unless hetero beats matched uniform baselines.

Provenance

```
doc/experiments/phase3/phase3_large_head_count_and_real_validation.md
results/phase3_toy_role_ontology_v2_large_heads_1000_20260523
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results/phase3_real_model_role_probe_pythia160m_deduped_float32_20260524
results/phase1_pythia160m_local_copy_candidate_pool_layers2_4_top2
results/phase1_pythia160m_wikitext103_natural_repeat_8gram_task_alignment_seed9_n128
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